

Referral radar dump

Referral Radar – AI-Powered Early Detection Engine

Referral Radar is an AI-powered early detection engine designed to support case managers by proactively identifying when and where to refer injured workers to appropriate healthcare programs. It uses natural language processing and predictive analytics to scan claim files, medical notes, and case histories, providing real-time, data-driven referral recommendations.

What is your vision for your idea?

Our vision is to create an intelligent, integrated referral system that transforms how case managers identify and act on early warning signs in claims. By embedding AI into the referral process, we aim to reduce long-term disability, improve recovery outcomes for workers, and ease the burden on staff and healthcare providers — ultimately creating a more responsive, efficient, and sustainable system.

What problem will your idea solve? Which stakeholders are affected?

The Problem:

Case managers can be overwhelmed by the volume and complexity of claims, leading to inconsistent, delayed, or missed referrals. This contributes to prolonged recovery times, increased costs, increased LTD, and staff burnout.

Affected Stakeholders:

- **Case Managers:** Struggle with manual, subjective decision-making under time pressure
- **Injured Workers:** Experience delays in receiving appropriate care
- **Healthcare Providers:** Receive mismatched or late referrals
- **Program Teams:** Face inefficiencies due to inappropriate referrals
- **Leadership:** Lacks visibility into referral trends and risk patterns

What are the biggest benefits of your idea?

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- **Time Savings:** Automates file review and referral suggestions
- **Cost Reduction:** Prevents claim escalation and reduces long-term disability costs
- **Risk Mitigation:** Identifies high-risk cases early for timely intervention
- **Improved Stakeholder Experience:** Enhances care coordination and satisfaction
- **Smarter Decision-Making:** Supports consistent, data-informed decisions and reduces staff burnout

Which categories apply to your idea?

Improve Processes and Stakeholder Experience

Enhance our Provider Network

Strengthening Success Metrics for our Programs and Services

Other

Team

1. Weicheng Xu – lead dev
2. Bilal Zubair - dev
3. Tyler Nguyen – coop dev. Intern dev
4. Dmitri Kiperman – solution architect
5. Aditya Chourasiya - QA lead
6. Ricky Kanwal - BE
7. Abhijit Dutta – BE
8. Kash Ajayi – scrum master
9. Jasper moedt - SME

The story1 works, we will start from there, as for the score and insights accuracy, this is always on going improvements, we are building our prototype for this one case for now, given we only have weekends and 2 days, 30th and 31st will be for presentation refinary and rehearsal, not realistic to do any programming with our 15inch monitor, We just want to have the UI ready (list view, details view) with the mind blowing map

, and the Agent trace view like this showing background multi-agent running,

then architecture design diagram to show where we deploy our app for production (example from other app, will get latest from Dmitri)

with these 3 concept, ppl knows this is ready, and other technical details we can work out after InnoFest, once we got funding.

Using chatgpt to replicate claim data/ conversations. Validated with case manager

Claim ID: #12345

Timeline: December 2023 – June 2024

 **Scene 1: Incident Report & Intake Call (Dec 18, 2023)**

Jennifer:

Hi, this is Jennifer from BlueRidge Insurance. I understand you're calling about a workplace injury — I'm here to help. Can you tell me what happened?

Sarah:

I fell down the stairs at work — about 30 steps — right in front of everyone. It was humiliating. My knee's swollen and I've been referred for scans.

Jennifer:

That must've been incredibly scary — and falling like that in a public place sounds really hard emotionally, too. Thank you for sharing that. Let's make sure we take care of both your physical and emotional well-being as we move forward.

 **Scene 2: Clinical Update – Imaging + Surgery (Jan 5, 2024)**

Sarah (email):

MRI showed torn meniscus. Surgery booked for Jan 20. I'm off work until then. Not sure how I'll handle going back after... everything.

Jennifer (reply):

Thanks for the update, Sarah. That's a big procedure — I hope you're getting rest. And I noticed your comment about returning to work — if that's weighing on you, I want you to know we can talk through that whenever you're ready.

 **Scene 3: Post-Surgery Recovery & Return to Work (Feb–Mar 2024)**

Jennifer (check-in call):

Hi Sarah, I'm checking in on how you've been doing since surgery. How's your recovery — physically and otherwise?

Sarah:

Physically I'm healing, yeah. But honestly, going back to work has been... weird. People keep looking at me. I feel like "the girl who fell." I've been taking side routes around the building just to avoid walking through the front.

Jennifer:

That sounds really tough. Feeling like you're under a spotlight can be emotionally exhausting — especially after an injury. Avoiding certain areas sounds like your way of coping. You're not alone in feeling that way. Has this been something you're dealing with daily?

 Scene 4: Mid-Claim Check-In (April 2024)

Jennifer:

Hi Sarah — how are things feeling now that you're a few weeks into being back at work? No pressure to talk only about progress — whatever's real for you is okay here.

Sarah:

My knee's fine, but I keep waking up dreading the walk into that building. My chest gets tight. I skip lunch with my team. I just don't want to be around people.

Jennifer:

Thanks for being honest. What you're describing — the tight chest, the dread, the isolation — those are important signals. A lot of people focus on physical healing, but what you're sharing tells me your recovery might still be happening — just in a different way.

 Scene 5: AI Insight Aggregation (May 2024)

Referral Radar Engine Output:

- “Humiliating fall” (intake notes)
- “Avoiding people,” “dread going in,” “tight chest” (calls)
- “Increased panic at work” (physician follow-up)

 Recommendation: Behavioral health referral suggested

 Scene 6: A Turning Point (May 18, 2024)

Jennifer:

Sarah, I've reviewed your history and I want to pause and acknowledge something: You've been incredibly strong through this process. But I'm also seeing signs that what you're experiencing might go beyond physical recovery.

Sarah: *(quiet)*

I... I guess I thought this was all part of getting back to normal.

Jennifer:

Totally understandable. But when you're waking up with dread, isolating yourself, and avoiding spaces because of anxiety — that's your body and mind trying to tell you something. And that matters just as much as a physical injury. I'd like to connect you with someone trained in helping people through this kind of post-trauma recovery. Would that feel okay?

Sarah:

Honestly? I think I need that.

 Scene 7: Mental Health Referral

Dr. Emily C. (initial assessment):

Hi Sarah, I've read over your file and spoken with Jennifer. What you're experiencing — the avoidance, the social panic — it aligns with what we sometimes see in post-incident trauma. It's real. And it's treatable.

Sarah:

It's kind of a relief to hear that.

 Scene 8: Claim Plan Revision

Jennifer (follow-up):

We've updated your plan to reflect mental health as a key part of your recovery. That means more time, more support, and more compassion — all of which you deserve.

 Scene 9: Introducing the Mental Health Program (Late May 2024)

Jennifer (call):

Hi Sarah, just checking in after your conversation with Dr. Emily. I wanted to let you know that we've gone ahead and enrolled you in our **Workplace Mental Health Recovery Pathway**. It's designed for people like you — recovering from both physical and emotional impact after a workplace event.

Sarah:

What does that involve exactly?

Jennifer:

It includes short-term sessions with a trauma-informed therapist, flexible coaching around workplace re-entry, and check-ins focused on confidence-building and anxiety reduction. It's fully covered, and designed to support people as they transition back — emotionally, not just physically.

Sarah:

That actually sounds really helpful. I've been feeling stuck.

Jennifer:

That's exactly what it's for. And I'll stay in touch to make sure things are working for you. We'll go at your pace.

 Scene 10: First Mental Health Check-In (Mid-June 2024)

Jennifer (email):

Hi Sarah,

Just wanted to check in — how's the first couple of sessions been? No pressure to share anything personal — I just want to make sure the support feels right.

Sarah (reply):

Honestly? I was skeptical. But the therapist helped me realize how much I was still carrying from the fall. I hadn't even said the word "embarrassed" out loud. Just naming it helped.

Jennifer (reply):

That's such a big step, Sarah. I'm proud of the progress you're making. Let me know if you ever want to tweak the plan — we're here to make this work for *you*.

 Scene 11: Reduced Symptoms and New Confidence (Early July 2024)

Jennifer (call):

Hi Sarah — I noticed your last note said you took the main stairs at work for the first time since the fall. That's a huge milestone. How did that feel?

Sarah:

Honestly? I was nervous — but I didn't feel like I had to hide anymore. It felt like I was reclaiming the space.

Jennifer:

That's powerful. You didn't just recover — you grew stronger. That deserves real recognition.

 Scene 12: Claim Closure Discussion (Mid-July 2024)

Jennifer (call):

Sarah, based on your progress — physically and emotionally — and the recommendation from your care team, we're ready to discuss closing your claim. That doesn't mean we stop caring. You'll still have access to support resources through your employer.

Sarah:

Wow. A few months ago I couldn't even make eye contact in the hallway. I didn't think I'd feel okay again. But I do. Thank you — for seeing the things I didn't know how to name.

Jennifer:

It's been a privilege to walk with you through this. This claim closes, but your resilience carries forward. I'm so glad you let us support your full recovery — not just the visible parts.

Lee, Cindy made some adjustments given your feedback. please review and let us know what you think. thanks!

the following is a second example

 Expanded Script: “Rob’s Recovery – Full Ecosystem View”

 Scene 1: Mike Calls Rob’s Primary Physician After Initial Injury (March 2019)

Mike (WorkSafeBC Case Manager)

Hi Dr. Jensen, it’s Mike calling from WorkSafeBC about Rob McCallister. He reported the logging truck incident earlier today.

Dr. Jensen (Physician):

Yes, I saw him at Merritt Hospital. He’s got a dislocated shoulder and multiple contusions. I’m recommending a follow-up MRI and orthopedic consult.

Mike:

Thanks. Any indication of psychological distress on your end?

Dr. Jensen:

He was stoic. Kept saying he’s “fine.” But I did notice he avoided eye contact when we talked about the collision. He mentioned the motorcyclist died.

Mike:

We’ll keep an eye on that. Let’s stay in touch — especially if any behavioural concerns emerge over time.

 Scene 2: Mike’s Internal Note — Suspicion of Delayed Trauma (July 2019)

Case Note (Mike):

“Rob cleared to return to work. Physical recovery is complete, but subtle signs of avoidance in language. No mental health supports accepted. Flagging for future psychological review if symptoms emerge.”

 Scene 3: Mike Reaches Out to Rob’s Employer (2025)

Mike:

Hi Carmen, this is Mike from WorkSafeBC. I’m calling about Rob McCallister — one of your long-haul drivers.

Carmen (Employer):

Oh, yeah. Rob’s been with us over a decade. Quiet guy. Keeps to himself.

Mike:

He’s reactivated his claim, citing delayed trauma from his 2019 accident. I wanted to check in on how he’s been performing at work.

Carmen:

To be honest... we've noticed changes. He's been late a few times — which is unusual for Rob. He takes odd detours on routes and avoids riding near motorcycles.

Mike:

Thanks for the honesty. We're connecting him to mental health support. It may be PTSD or unresolved trauma. Is your team open to a gradual return-to-work if needed?

Carmen:

Of course. Let us know how we can support — we just want him safe and well.

 Scene 4: Mike Consults With Psychological Advisor (May 2025)

Mike:

Hi Dr. Mahdi, thanks for taking my call. I've referred Rob McCallister to your trauma program.

Dr. Mahdi (Clinical Psychologist):

Yes, we've completed the intake. Rob's showing classic signs of delayed-onset PTSD — hypervigilance, intrusive imagery, emotional numbness.

Mike:

Any concerns I should be watching for on the case management side?

Dr. Mahdi:

He struggles with stigma — very internalized. If his return to work feels rushed or if he senses judgment from coworkers, that could derail progress. Reintegration needs to feel safe.

Mike:

Understood. I'll work with his employer to plan something trauma-informed. Please keep me posted on milestones.

Dr. Mahdi:

Will do. He's already showing willingness — which is promising.

 Scene 5: Mike's Return-to-Work Planning Call With Employer (September 2025)

Mike:

Carmen, I wanted to loop back on Rob's progress. He's stabilized significantly in therapy and we're prepping for return-to-work.

Carmen:

That's great news.

Mike:

I'd recommend a gradual re-entry. Fewer hours to start. Possibly avoid high-traffic highway routes for the first month.

Carmen:

Noted. I'll have our RTW coordinator set that up. Do we mention anything to the other drivers?

Mike:

Just that Rob's returning on a modified schedule, and that we're supporting his full recovery. We want to avoid any whispers — let him ease in.

Carmen:

Got it. He deserves that.

 Scene 6: Mike Logs Claim Closure After Final Check-In (October 2025)

Case Note (Mike):

“Spoke to Rob post-return. Reports improved sleep, emotional clarity, and confidence. Employer satisfied. Psychological treatment ongoing but stabilized. Claim officially closed. Rob thanked the team for ‘not forgetting about him.’”

Prompts for the above

Did vectorization of the policies/ business rules

The agent then starts a conversation with LLM

LLM makes recommendation

UI flow

Built 3 MCP servers for

- Claim info

- Business rules and policies
- Healthcare programs

The above is one azure service with multiple MCP servers

Claims doc are in raw format, no vectorization is done

Docs are already converted to text format (from pictures etc)

Agentic process

We want to have multiple agents working together (multi agentic, multi turnaround scenario)

Another idea we are trying to sell is multiple MCP servers

Can be done in one shot too if we have a powerful LLM (like have one master agent that assigns to mcp server) but for POC we can just do smaller scale and use one master agent that is trying to orchestrate between a claims agent and business rules

Architecture says 4.1 but microsoft suggests 01 (for reasoning the policy)

We have an MCP server against sharepoint for policy

We want to sell the idea of business domain specific mcp servers as our scale is pretty large

Lang graph and lang chain we are using for orchestration

Microsoft suggests an iterative approach where MCP servers are wrapped around agents.

To scale, scale on MCP server side

What do we mean by scalability – larger number of documents

Whats the best practice to host langchain/lang graph agents handling multiple claims and run them in parallel - AKS

When a recommender agent is trying to analyze policies given one claim information, it is not able to do - multi-turn discussion between agents

Termination condition needs to be met to end conversation – semantic kernel (library)

Can we publish our agent and mcp to azure foundry so that it is visible to others in the org? We want to share this with teams to leverage and repurpose

Can always create an endpoint and expose

Sharepoint ingestion (vectorization) is in beta (in preview)

Multi-agent multi-turn – there are 2 agents one asks questions, second responds, agent 1 has validation capability too. All gets stored in the memory. Agent 1 can rephrase with more info if it is not happy with the answer

Say agent 1 asks for relevant policies for a claim that is both trauma and mental health. But agent 2 gives only trauma policies. So multi-turn is agent 1 asking agent 2 to keep going till it returns mental health only policies. Until you don't get mental health policies

What is the extra work required to better the validation agent -

What are you trying to optimize for? This part agent cannot add. We have to add.

Like – check for coverage issues, check for feasibility

Because this is agentic, a lot of memory and context is generated automatically. Do we have persistent storage for everything it generates

For example, the business rules agent may learn from other conversations maybe add space in db to for it (for pre-processing)

We can also agentic rag, graph rag – these aren't giving right results from policy data

Policy is not changing often, it is static. Can we do semantic chunking instead. So we are not extracting info sequentially but can we instead extract something more concrete/ rule specific content – some kind of pre-processing on the policy so that agent can understand what we are asking. Right now it cannot.

This can help with

- Multi-turn
- And terminating condition

There are 4 agents currently that are dealing with policy

Q-should we build a custom chatbot or integrate it in teams? Depends on how the user interacts right now

Main agent can have data security routing. Get the main agent to get the identity of the user.

Portfolio Template (Hackathon / POC Version)

1 Project Snapshot (Now More Concise + Impactful)

Project: Referral Radar – AI Mental Health Referral Assistant

Context: 2-week innovation hackathon (2nd place out of X teams)

Role: Product Lead

Team: X engineers, X data scientist, X SME clinician

Scope: Built production-ready POC with live AI agent, recommender logic, and MCP-based architecture

Goal: Validate whether agentic AI could improve mental health referral decision support in claims workflows.

2 What Makes This Strong (In Tech Terms)

This is now about:

- AI feasibility validation
- Fast prototyping
- Strategic framing
- MVP scoping under extreme time constraints
- Communicating vision clearly to execs

That is VERY 0→1 startup-style PM.

3 What You Must Highlight Differently

A. 0 → 1 Framing

Answer:

- How did you identify this problem?
- Why was this worth building?
- What signal convinced you AI could solve it?

Tech companies want to see **insight generation**, not just execution.

B. How You Scoped the MVP in 2 Weeks

This is critical.

Explain:

What you cut:

- Full model retraining
- Production governance
- Large-scale integration

What you prioritized:

- Extract + summarize
- Program recommendation logic
- Explainability
- Chat interface
- MCP server abstraction

This shows product judgment.

C. AI Product Thinking

Instead of production metrics, focus on:

- SME alignment testing (80% agreement target)
- Example case walkthroughs
- Agent trace demonstrations
- Explainability UI
- Architecture extensibility

You are showing:

“I understand how AI systems should be built responsibly and scalably.”

D. Demo & Storytelling (Underrated Skill)

Hackathons are judged on:

- Clarity
- Narrative
- Business alignment
- Strategic thinking

If you placed 2nd, that means:

- Your framing resonated
- The business value was clear
- You communicated impact well

That is a big deal.

Add:

- What feedback judges gave
- Why you think it ranked highly
- What differentiated it

[Decisions I Drove](#)

[Artifacts](#)

[Innofest presentation](#)

[Link](#)

Problem framing / problem statements considered and prioritized

Problem statements:

Problem n: Reasoning for Claims engine prediction

Empower **Claim owners** with clear reasoning behind the Claim Priority/Risk Score by leveraging Claim Prediction Engine (*show them HOW the model reached the score*)

Business Value:

1. Confidently adopt Claim Prediction Engine to prioritize claims supported by clear data
2. Provide additional supplementary information on *How the score was reached*

Success Metrics:

1.

Problem 2: Recommend best health care program

Help Claim owners identify the most suitable referral for MH claims using Agentic AI to guide decision support, via natural language.

Business Value:

- Clarify claim context from overlapping services
- Surface most relevant healthcare programs for the specific claim

Success Metrics: [@Dutta, Abhijit](#) **** [@Dodani, Khushboo](#) ****

1. Agent AI consistently recommends healthcare programs aligned with claim owner decisions or approved guidelines (e.g., 80%+ agreement rate with SME-reviewed cases).
2. Low false positive/negative rate in recommendations, minimizing inappropriate referrals.

3. Agentic AI accurately parses and summarizes claim context, distinguishing between overlapping services and identifying key factors (e.g., chronicity, comorbidities, prior program use).
4. Agentic AI can extract relevant unstructured data from medical notes, history, and case narratives.
5. Claim owners can interact with the AI via simple, natural language prompts (e.g., “What are the best options for this claim?”).
6. System responds in clear, actionable language, reducing training time for claim owners.
7. Measure reduction in time spent by claim owners in researching referral options (e.g., X% faster decision-making time).
8. Positive user feedback on AI reducing cognitive load.
9. Seamless integration with claim systems (CMS or equivalents) without disrupting current workflows.
10. Easy access via existing claim management tools or portals.
11. Improved claim processing efficiency (e.g., reduced open duration for MH claims).
12. Potential reduction in repeat referrals or inappropriate program placements.
13. Better claimant outcomes tracked via downstream metrics (e.g., reduced time on claim, improved return-to-work rates).

Problem 3: On demand chat based access to Claims data

Enable Claim owners to chat with an agent for real-time clarification on complex claim details

Business Value:

Accelerate claim resolution by reducing back-and-forth and improving understanding through instant, guided support

Problem 4: Democratized access to Claim information for future Agentic projects

Introduce a claim specific MCP server using Microsoft Azure MCP service that will introduce context rich access to most public and private

Elevator Pitch: Smarter Referrals, Stronger Recovery

Referral Radar is your intelligent assistant — scanning claim data for hidden mental health signals and helping case managers make faster, more confident decisions

OPEN Questions

1. Can we identify missing documents that could further help improve the healthcare program recommendation so that the claims owner can ask for these specific documents and intelligence ?

2. C.....

Problem

1. **Problem Statement:** The team must provide a clear and concise description of the problem the idea aims to solve. Is it well-defined, including target stakeholders and a quantified impact? Is it worth addressing?

Claim owners face two key barriers to making timely, appropriate mental health (MH) referrals:

1. **Lack of decision support:** Critical context—such as red flags, comorbidities, and prior history—is buried in unstructured data, making it difficult to assess referral needs quickly. The goal is to identify mental health referrals earlier on to guide early intervention.
 2. Insights, Summary
 3. Chronological display of documentation
 4. Mind map, Chatbot?
 5. **Unclear program fit:** Even when MH needs are identified, overlapping services and inconsistent guidance make it hard to match claims to the most appropriate healthcare programs—leading to delays, repeat referrals, and higher costs. The goal is to identify appropriate mental health programs for injured workers.
- Can we predict accurately with the existing mental health options
 - Business vision - Scale from 4 programs to X programs,

Product

1. **Product Desirability:** Did the team get user feedback on their concept? Did the team demonstrate that users show a strong desire for this product?
2. **Product Maturity:** How well did the team create a Minimum Viable Product (MVP) with functional features? Consider the design for usability, the quality and completeness of the features.

Product

Referral Radar is your intelligent assistant — scanning claim data for hidden mental health signals and helping case managers make faster, more confident recommendations

Features

1. Claims ****AI agent (user facing)
2. **Extractor** Agentic AI can extract relevant unstructured data from medical notes, history, and case narratives.
3. **Parser** Agentic AI accurately parses and summarizes claim context, distinguishing between overlapping services and identifying key factors (e.g., chronicity, comorbidities, prior to program use).

4. **Chat** agent that enables case managers to chat with an agent for real-time clarification on complex claim details
5. **Recommender** Agent AI consistently recommends healthcare programs aligned with claim owner decisions or approved guidelines (e.g., 80%+ agreement rate with SME-reviewed cases).
6. **Seamless integration** with claim systems (CMS or equivalents) without disrupting current workflows.
7. MCP server (technical advancement)
8. **Claims and Procedure MCP:** The MCP (Model Context Protocol) server acts as a domain-specific middleware layer that consolidates APIs and document resources related to claims, enabling agentic applications to access claim data efficiently without custom integrations for each agent.
9. **Scalability and Extensibility :** It provides a standardized interface for agents to retrieve, process, and reason over claim-related information, supporting multi-agent frameworks and future extensibility. 2
10. **Security and Governance:** The MCP server enhances security and governance by centralizing access control, allowing fine-grained permissions for different teams or agents.

Impact

1. Value Proposition: Did the team clearly outline what benefits or improvements the idea/solution will deliver? Was the value proposition quantified? Is the value substantial enough to justify investment? (ex. Does the idea move the needle in optimizing our internal ITAP operations?)
2. Strategic Alignment: Is the idea aligned with divisional goals and strategic priorities?

Impact statements

1. Reduction in average time to make referral decisions (parsing information)
2. Decrease in open claim duration for MH cases
3. Reduction in inappropriate or repeat referrals
4. CO satisfaction/ trust score for recommended programs
5. Reduced workload for clinical advisors or escalation teams
6. Days between claim acceptance and first referral
7. RTW rates

Implementation

1. Implementation Readiness: How close is the presented product ready for implementation?

2. Operationalizing: How well did the team highlight next steps for their product in terms of implementation and future features? Did the team articulate what is required to operationalize their idea (ex. ongoing costs for the solution, sustainment and maintenance)?

Future features

1. Identify missing documents that could further help improve the healthcare program recommendations.
2. Empower case managers ****with clear reasoning behind the Claim Priority/Risk Score by leveraging Claims Prediction Engine (*show them HOW the model reached the score*). Our hypothesis is that this will help mitigate risks of adoption.

Roadblocks to implementation

Presentation

1. Presentation Quality: Evaluate how effectively the team presented their idea. Was it engaging, creative, and memorable? Consider the clarity of their message, visual aids, and overall storytelling.
2. Q&A Quality: Assess how well the team handled the Q&A session. Did they address questions confidently and comprehensively?

Slide Flow:

Intro and Problem statement

MH stats/MH policy and outlook as a societal trend

Use of claim cost duration

CM workflow and numbers (300 CM up to 200 files on desk)- what numbers are missing- (how long does a claim review take? How much time are we saving them?)

Outcome (longer tx/higher cost/poor provider and worker experience) Add stats on early tx and outcomes

(for myself coming at it from a clinician perspective I have years of seeing this in place etc..)

Solution

Building on existing work from Diego(name of that tool?)

Impact

Building on existing workflow and work being done

Early identification/more accurate referrals to hcp

FTE savings in CM times(rough calculation)

Ease of CM role

Keeping people work for people(CM feedback note) reduced admin burden

Reduced claims cost as improved referral timelines

Scaleability to all conditions and referrals organizational impact

Stats:

- Problem statement - 1 min (jasper)
- Product desirability - 1 min (jasper)
- Impact (jasper) - 2 mins
- strategic alignment
- quantifiable business value
- Product & implementation (bilal) - 4mins
- UI
- Agent trace
- Architecture & implementation readiness
- Value proposition round up - 1 min (khush)
- Implementation readiness
- Operationalizing/ next steps - 1 min (aditya)

[Referral radar Points.pdf](#)

- chose mental health referrals as a program (out of 78 healthcare programs at worksafebc) to build the POC on

Mental Health as a Growing Crisis: Consolidated Summary

1. Psychological Injury Claims Trends (YTD June 2025)

Key Points:

Psychological injury-only claims are rising steadily, both in volume and complexity.

Claims are increasingly being accepted under Section 135 of the Workers Compensation Act.

Concurrent physical and psychological injuries are also increasing, indicating compounding trauma.

Top sectors impacted include healthcare, education, and public administration.

2. End-to-End Mental Health Treatment Review (2021–2024)

Key Points:

Thousands of referrals annually across multiple treatment streams (RSS, Standard, TSS, SS, SER).

PTSD claims require the most intensive care: average of 5.1 blocks and 34 sessions.

Return-to-work (RTW) is often delayed, with many workers needing multiple treatment blocks.

Systemic issues include fragmented care pathways, limited integration, over-reliance on contracted services, and rising costs (e.g., Standard Treatment > \$6M/year).

3. WorkSafeBC Mental Health Strategy (2023–2025)

Key Points:

Psychological injury claims increased 118% between 2018 and 2022.

\$6.6 billion annual cost to B.C.'s economy due to poor mental health.

70% of Canadian employees are concerned about psychological safety at work.

Strategic focus areas include:

- Prevention (e.g., new regulations, risk assessment tools)
- Engagement (e.g., outreach, consultations)
- Partnerships (e.g., with providers and agencies)
- Evidence-informed treatment
- Leadership (WorkSafeBC modeling best practices internally)

4. Unlocking the Future: Global Scan (Q2 2025)

Mental Health-Relevant Insights:

Mental health is declining globally, especially among young workers.

Top stressors include workplace conflict, burnout, and performance pressure.

Jurisdictions are responding with right-to-disconnect policies, expanded sick leave, and 4-day work week trials.

Unconventional therapies (e.g., psilocybin, ketamine, neuromodulation) are gaining traction.

Bullying, harassment, and violence are increasing across sectors, with links to depression and suicidality.

Standard Mental Health Treatment – Summary of Key Metrics

[Shape](#)

Utilization Metrics (Mental Health Claims)

Admission-Level Data:

- Average sessions per admission: 15.8

- Median sessions per admission: 11
- Percentage receiving more than one block: 55%
- Average sessions per block: 6.4
- Average blocks per admission: 2.5
- Extension reasons:
- 3,717 cases exceeded time limits
- 12 cases exceeded session limits

Claim-Level Data (n = 2,023):

- 31% received one block
- 20% received two blocks
- 16% received three blocks
- 10% received four blocks
- 22% received five or more blocks
- Average blocks per claim: 3.1
- Average sessions per claim: 19.7

[Shape](#)

Financial Overview

- Annual referral costs:
- 2021: \$5.74 million
- 2022: \$5.83 million
- 2023: \$6.78 million
- 2024 (YTD): \$6.16 million

[Shape](#)

Return to Work (RTW) Outcomes

- Average blocks to RTW/mRTW: 1.6
- RTW occurs:
- In the first block: 71% of cases
- In the second block: 12%
- In the third block: 7%

- In the fourth block: 5%
- Only 37% of first blocks result in RTW/mRTW
- use and leverage existing tools and work such as the claims prediction engine. we had a goal to win innofest and we had to hack in 2 days. time as a constraint, found models already in production and doing well to build on top of

[PQSS Meeting - Claim Prediction Engine Overview 1.pdf](#)

[ClaimPredictionEngine_README_2025Mar14.html](#)

demo recording

2 System Architecture Diagram

Simple:

- Claims data
- AI model
- Provider view
- Feedback loop

Signals technical literacy.

3 Problem Framing Slide

Before solution:

- Referral bottlenecks
- Manual distribution inefficiency
- Hypothesis: predictive matching reduces delays

Big Tech values problem framing over solution glamour.

4 Hackathon Constraints Slide

- 2 weeks
- Limited data
- Rapid prototype
- Team structure

Shows bias toward action.

5 “If Taken to Production” Roadmap

- Data validation
- Model refinement
- A/B testing
- Governance review

Shows product maturity beyond hackathon.

can use the following for visualization on portfolio website

[Referral Radar - IS What's Cooking.pdf](#)